

Ideation & Problem Validation

Submission 1 · FlowState: The Autonomous Academic Project Manager

TEAM NAME

Mutex Minds

MEMBERS

- Kirtan – Testing and Development
- Khanjan – Backend
- Konark –DB and DevOps



Problem Statement

Core Problem (1–2 sentences)

Large academic project teams suffer from collaboration friction, unequal workload distribution, and missed dependencies because traditional coordination requires too much manual effort.

Context

University environments, specifically medium-to-large engineering or research project groups.

Scale

Virtually every university student working in teams of four or more members faces this coordination breakdown.

Evidence

Personal lived experience of team members constantly chasing others for updates on WhatsApp or Discord, leading to last-minute crunch times and burnout.

Tip: A strong problem statement answers WHO suffers, WHAT they can't do, and WHY current solutions fall short.

Target Users

Primary User Persona

Who are they?

University Students

Age / Context

18 - 24

Key pain point

The "I thought you were doing that" syndrome and unequal work distribution causing deadline panic.

Current workaround

How do they cope today?

Why This Problem Matters

Scale / Reach

Occurs in almost every single group project assigned globally every semester.

Economic Impact

Significant student time wasted on logistics instead of academic research, leading to lower grades and severe academic burnout.

Social / SDG Fit

SDG 4 (Quality Education) – ensuring equitable learning environments and accurate, fair assessments.

Urgency

Project sizes and technical complexity in modern coursework are increasing, making manual management mathematically impossible for students to juggle.

Tip: Use numbers wherever possible. Even rough estimates are better than vague claims.

Existing Solutions & Their Limitations

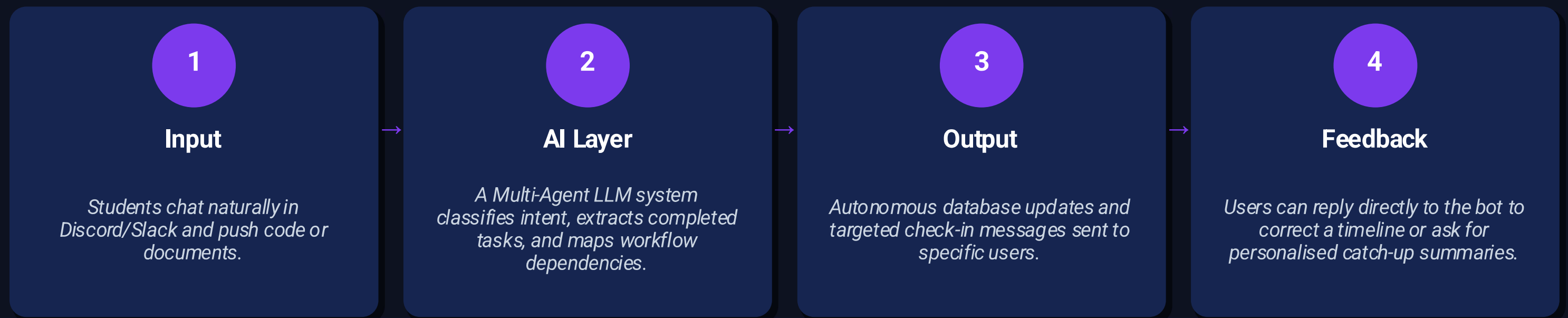
Existing Solution	What it does	Key Limitation / Gap
Solution (WhatsApp/Discord)	<i>General team communication</i>	<i>No task automation; critical information and deadlines get buried in chat.</i>
Solution 2 (Notion/Trello)	<i>Manual task tracking boards</i>	<i>Requires constant manual updating, which students inevitably abandon mid-semester.</i>
Solution 3 (In-person Syncs)	<i>Weekly progress meetings</i>	<i>Slow, unscalable, and usually unproductive without a structured agenda.</i>
The Gap We're Targeting:	<i>None of the existing solutions autonomously track progress. They all require students to do the manual labor of project management, which we will eliminate using ambient AI tracking.</i>	

Proposed Solution Overview

Your Idea in One Line

We are building an autonomous AI project manager for university student groups that helps them track dependencies and workloads using ambient chat extraction, unlike existing solutions which require manual data entry.

HOW IT WORKS



Tentative Feature List

Must Have (MVP)	Should Have	Nice to Have
Ambient Discord tracking bot	Procrastination & momentum engine	Smart RAG catch-up briefings
LLM text-to-status parser	GitHub push webhook listener	Visual task momentum dashboard
PostgreSQL task dependency logic	Automated private Discord nudges	Multi-agent AI task delegation

💡 *Tip: Be realistic about what you can build in 7 days. A focused MVP beats a feature-bloated prototype.*

Tentative Tech Stack & AI Tools

AI / ML Layer

Gemini API (intent classification), LangChain (multi-agent orchestration), and Ollama running Llama-3 for zero-cost local testing

Frontend

Next.js (React) for the analytics dashboard, integrated alongside the Native Discord API for chat-based user interactions.

Backend

Python FastAPI to handle high-concurrency GitHub webhooks, Discord event listeners, and REST API routing.

Database

PostgreSQL for strict relational state management (Users/Tasks/Logs), combined with QdrantDB for chat history embeddings

Hosting / DevOps

Docker for local containerization, Vercel for instant frontend hosting, and Render for the API backend services

Expected Impact

1 Lakh +

Users Impacted

estimated reach after launch

50%

Efficiency Gain

time / cost saved per user

Z

Key Outcome Metric

your chosen north-star metric

Short-term Impact (3–6 months)

Eradicates team "ghosting" by providing early procrastination warnings, letting students focus on actual academic work rather than logistical management

Long-term Vision (1–2 years)

Adoption by university LMS systems to give professors transparent, empirical data on individual contributions for fair group grading.

Thank you — we're excited to build this!